

PA Club Heartitude · Panamá 2026

Mathematics for Panama: A Tool Born from the Classroom



Free digital platforms · Based on MEDUCA 2024 · Preschool and Elementary Grades 1–6

Jiyun Kim

Instagram: @heartitude_ba

THE STARTING POINT · PANAMA, 2021

I arrived in Panama. And something caught my attention.

At a café, I paid for my coffee and received change. **Something didn't add up.** I counted again — it was wrong.

At the supermarket, the same thing happened. Sometimes **too little**, sometimes **too much**. It wasn't dishonesty — there was something deeper going on.

Could this be related to how basic arithmetic operations are taught here?

A SINCERE QUESTION

I have always loved mathematics.
So I asked myself:
Is math taught the same way everywhere?



I reviewed the official curricula of all three countries and compared them in detail.

WHAT I FOUND

Mathematical content is universal

All three curricula cover essentially the same topics

Grade	KR Korea	US USA	PA MEDUCA
Grades 1–2	Numbers, addition & subtraction, intro to multiplication	Basic operations, place value, measurement	Number recognition, basic operations, shapes
Grades 3–4	Fractions, decimals, multiplication & division	Mult. & div., fractions, area & perimeter	Mult. & div., fractions, decimals, measurement
Grades 5–6	Ratios, statistics, probability	Ratios, statistics, introductory algebra	Advanced fractions, ratios, statistics, intro algebra

Equivalent content: The MEDUCA curriculum is aligned with international standards

THE PROBLEM I IDENTIFIED

The previous curriculum moved **too fast**

Accelerated progression

Some content was introduced in very early grades, before students could properly absorb it.

No time to consolidate

Students moved to the next topic before the previous one was **fully understood and practiced**.

Fragile foundation

Without mastery of the four basic operations, everything that follows — fractions, ratios, algebra — becomes much harder.

The reform improved the pace. **But gaps remain.**

What improved

The progression was adjusted. The **Fundamental Learning Rights (DFA)** approach was adopted — comprehension first, then advancement.

What still needs work

Some content is still introduced **earlier than in Korea or the US** — and for those topics, there are insufficient practice tools.

The opportunity

The reform is heading in the right direction. What it needs now are **digital tools** that support that pedagogical vision.

THREE SOURCES · ONE CONCLUSION

This need didn't come from a book.

Personal observation: In cafés and supermarkets, change calculations were frequently incorrect — a sign of fragility in basic arithmetic operations.

Voices from the field: Volunteer teachers and parents confirmed: children need **more practice with the four basic operations** from an early age.

Curriculum analysis: Comparing three national curricula confirmed the gap in practice tools adapted to the Panamanian context.

All three sources pointed to the same place: basic operations · early practice · accessible tools

THE ANSWER

Two platforms · One philosophy

If the foundation is strong from an early age, everything built on top of it is more solid.

Mathematics for Preschool

Basic operations from an early age — through **play**. Because if numbers feel fun from the start, the path ahead becomes much easier.

- Number recognition
- Addition & subtraction
- Times table bingo
- Medals & rewards
- Mobile-optimized

latidos-ninos.vercel.app

Mathematics for Elementary · Grades 1–6

Based on the **Fundamental Learning Rights (DFA) · MEDUCA 2024**. So every student can understand, practice, and move forward at their own pace.

- Learn
- Vocabulary
- Practice
- Evaluate
- Game Zone

jy-matematica-panama.vercel.app

PRIMARY PLATFORM

Mathematics for Elementary School

Developed in alignment with the **Fundamental Learning Rights (DFA)** of MEDUCA, based on the Methodological Guide for Mathematics · 2024 curriculum revision.

- DFA · MEDUCA 2024
- Free
- Grades 1–6
- In Spanish
- No installation needed

jy-matematica-panama.vercel.app

Learn (Aprende)

Clear conceptual explanations with visual examples adapted to the Panamanian context.

Vocabulary (Vocabulario)

Precise mathematical glossary for each grade and unit.

Practice (Practica)

Interactive exercises with immediate feedback — enough time to truly consolidate.

Evaluate (Evalúa)

Unit-level achievement measurement aligned with DFA indicators.

Game Zone

Speed Blitz · Drill Clásico · Memory Match — playful reinforcement.

CURRICULUM ALIGNMENT

PA Based on DFA • MEDUCA 2024

Content, competencies, and achievement indicators · Grades 1–6 of Elementary School

GRADE 1**Foundations**

- Numbers 0–100
- Basic addition & subtraction
- Geometric shapes
- Length measurement

GRADE 2**Building**

- Numbers up to 1,000
- Multiplication tables
- Simple fractions
- Time & calendar

GRADE 3**Expanding**

- Multiplication & division
- Numbers up to 10,000
- Equivalent fractions
- Perimeter & area

GRADE 4**Deepening**

- Decimals
- Fractions: add & subtract
- Angles & polygons
- Basic statistics

GRADE 5**Applying**

- Operations with decimals
- Ratios & proportions
- Percentages
- Volume

GRADE 6**Integrating**

- Introductory algebra
- Statistics & probability
- Advanced ratios
- Spatial geometry

A SINCERE REQUEST

I built the tool. **You have the pedagogical knowledge.**

I'm a student, not an education expert. That's exactly why your perspective as future teachers means so much more than mine. Would you help me make it better?

01

Try it out

Use it freely. What feels right? What feels incomplete or off — from the perspective of someone studying to teach?

02

Evaluate it

Are the content and progression pedagogically sound? You are training to know this far better than I do.

03

Tell me what's missing

Any observation, however small, would help me enormously in making this a better resource for Panamanian students.

THANK YOU

**Let's build together
the mathematical future
of Panama's children.**

This is only the beginning. We hope this is the start of a genuine, lasting, and meaningful collaboration.

ELEMENTARY · GRADES 1-6
jy-matematica-panama.vercel.app

PRESCHOOL
latidos-ninos.vercel.app

jyun.k.heartitude@gmail.com Instagram: @heartitude_ba